

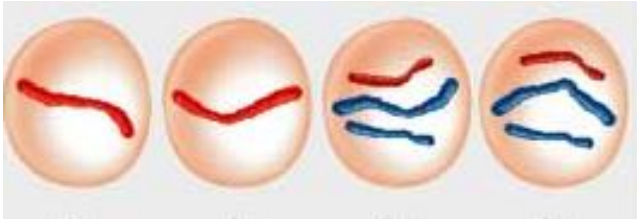
MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose/write the correct answer for the following questions. 20

- 1. Paternal chromosome deletion in mammals can lead to many physiological abnormalities, such as Prader Willi Syndrome, this can be explained through**
(i) RNA interference (ii) Genomic imprinting
(iii) Epistasis (iv) All of above
- 2. The insertion sequence that are involved in horizontal gene transfer through conjugation is functionally similar to**
(i) Transposon (ii) Inverted repeats
(iii) Change in orientation of small chromosomal segment (iv) Wild type host
- 3. When a pure lines of white flower plant crossed to red flower plants, in F₂ three different phenotypes of flowers were observed, the phenomenon can be explained through**
(i) Co-dominance. (iii) Epistasis.
(ii) Incomplete dominance. (iv) None of the above
- 4. Fungi in the class Ascomycetes are helpful for the study of meiosis. Which of the below statements is false?.**
(i) Each ascospore has 2 chromatids.
(ii) The ascospores in an ascus contain all four of the chromatids from a tetrad.
(iii) The segregation of independent mutations can be studied.
(iv) Fungi are easy to propagate.

5. "The severity of the disease condition is dependent on the number of disabled mitochondria present in the egg". This statement relates to
 (i) Co-dominance (ii) Organelle inheritance (iii) Sex-linkage (iv) Paternal inheritance
6. The gene for pili formations is present on which region of F-Plasmid?
 (i) Tra region (ii) Mob region (iii) Ori C (iv) Tn 1000
7. PCR based DNA amplification is not an essential feature of which of the following molecular marker?
 (i) RFLP (ii) RAPD (iii) SSR (iv) ISSR
8. Find the correct expression for variance
 (i) $\Sigma[(X_i - \bar{X})(Y_i - \bar{Y})]/(n-1)S_X S_Y$ (ii) $\Sigma[(X_i - \bar{X})(Y_i - \bar{Y})]/(n-1)$ (iii) $\Sigma(X_i - \bar{X})/(n-1)$ (iv) $\Sigma X_i/n$
9. 
- Look to above image, and identify the genetic anomalies associated with cell division.
 (i) Meiosis (ii) Gene deletion (iii) Non-disjunction (iv) None
10. Somatic cell hybrids can be used for
 (i) Construction of gene library
 (ii) For locating a specific gene on chromosome
 (iii) For construction of chromosome map
 (iv) All of above
11. "Phage Mu is temperate phage "
 (i) True (ii) False (iii) Cannot be said (iv) True in certain conditions
12. Lambda phage can integrate itself by
 (i) By replicative transposition (iii) Site specific insertion
 (ii) By cut and pest transposition (iv) None of above
13. If we cross 2 plants that are heterozygous for four recessive alleles, what fraction of the progeny will be homozygous for all recessive alleles?
 (i) 1/16 (ii) 1/128 (iii) 1/256 (iv) 1/200
14. Give the mode of inheritance for the given pedigree_____
- 
15. From the following which represents the non-crossing over product of tetrad?
 (i) DDdddddD (ii) DDddDDdd (iii) ddddDDDD (iv) ddDDddDD
16. $V_P = V_G +$ _____
 (i) V_A (ii) V_D (iii) V_E (iv) V_i
17. $H^2 = V_G/V_P$ [True] / [False]

18. RFLPs or restriction fragment length polymorphisms can be defined by:

- | | |
|---|--|
| (i) More than one nucleotide substitution in a DNA fragment. | (ii) A single nucleotide substitution in a restriction enzyme site in a DNA fragment. |
| (iii) A single nucleotide substitution in a restriction enzyme site in a gene. | (iv) None of above |

19. Some changes are encountered in the infectious particle as it infects the cell. Choose the incorrect statement.

- (i)** The infectious particle circularizes upon causing infection by annealing at cos sites
- (ii)** There are two promoters present, PR and PL
- (iii)** Upon infection only PR is activated
- (iv)** Promoters give rise to immediate early transcripts

20. Phase variation in flagella antigenic structure can be observed in *E.coli* [True]/[false]

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IInd Semester of MSc Biotechnology/Biochemistry Examination-May-2019

BT762: Molecular Genetics

Date: 04/05/2019 Day: Saturday Time: 10.30 a.m. To 1.00 p.m. Maximum Marks: 50

Instructions:

- Section I and II must be attempted in TWO ANSWER SHEET.**
- Make suitable assumptions and draw neat figures wherever required.
- Use of non-programmable calculator is allowed.
- Show necessary calculations.

SECTION -I

Q-IIA

Answer the following questions as directed [Any Two]

8

1

Cross	Repulsion	Progeny		Percent Recombination
	Heterozygote	Parental	Recombinant	
1	<i>bw+ / + c</i>	850	503	37.2
2	<i>bw+ / + Blk</i>	750	237	24.0
3	<i>c+ / + Blk</i>	629	183	22.5

Are any of the three genes studied in above crosses are linked? Show the relative position of these three genes on linkage map.

2

An F⁺ strain, marked at 10 loci, gives rise spontaneously to Hfr progeny whenever the F factor becomes incorporated into the chromosome of the F⁺ strain. The F factor can integrate into the circular chromosome at many points, so that the resulting Hfr strains transfer the genetic markers in different orders. For any Hfr strain, the order of markers entering a recipient cell can be determined by interrupted mating experiments. From the following data for several Hfr strains derived from the same F⁺ strain, determine the order of markers in the F⁺ strain.

Hfr Strain	Markers Donated in Order
1	— Z-H-E-R →
2	— O-K-S-R →
3	— K-O-W-I →
4	— Z-T-I-W →
5	— H-Z-T-I →

3	Bacteriophage P1 mediates generalized transduction in <i>E. coli</i>. A P1 transducing lysate was prepared by growing P1 phage on <i>pur</i>⁺ <i>pro</i>⁻ <i>bis</i>⁻ bacteria. Genes <i>pur</i>, <i>pro</i>, and <i>bis</i> encode enzymes required for the synthesis of purines, proline, and histidine, respectively. The phage and transducing particles in this lysate were then allowed to infect <i>pur</i>⁻ <i>pro</i>⁺ <i>bis</i>⁺ cells. After incubating the infected bacteria for a period of time sufficient to allow transduction to occur, they were plated on minimal medium supplemented with proline and histidine, but no purines to select for <i>pur</i>⁺ transductants. The <i>pur</i>⁺ colonies were then transferred to minimal medium with and without proline and with and without histidine to determine the frequencies of each of the outside markers. Given the following results, what is the order of the three genes on the <i>E. coli</i> chromosome? <table><tr><th>Genotype</th><th>Number Observed</th></tr><tr><td><i>pro</i>⁺ <i>bis</i>⁺</td><td>100</td></tr><tr><td><i>pro</i>⁻ <i>bis</i>⁺</td><td>22</td></tr><tr><td><i>pro</i>⁺ <i>bis</i>⁻</td><td>150</td></tr><tr><td><i>pro</i>⁻ <i>bis</i>⁻</td><td>1</td></tr></table>	Genotype	Number Observed	<i>pro</i> ⁺ <i>bis</i> ⁺	100	<i>pro</i> ⁻ <i>bis</i> ⁺	22	<i>pro</i> ⁺ <i>bis</i> ⁻	150	<i>pro</i> ⁻ <i>bis</i> ⁻	1																		
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Q-IIB	Answer the following [Any one]	4																											
1	A researcher has discovered a new blood-typing system for humans. The system involves two antigens, P and Q, each determined by a different allele of a gene named N. The alleles for these antigens are about equally frequent in the general population. If the N_P and N_Q alleles are codominant, what antigens should be detected in the blood of NP NQ heterozygotes?																												
2	Flower color in a garden plant is under the control of a gene with multiple alleles. The phenotypes of the homozygotes and heterozygotes of this gene are as follows: <table><tr><th colspan="3">Homozygotes</th></tr><tr><td><i>WW</i></td><td>red</td><td></td></tr><tr><td><i>ww</i></td><td>pure white</td><td></td></tr><tr><td><i>w^sw^s</i></td><td>white stippled with red</td><td></td></tr><tr><td><i>w^pw^p</i></td><td>white with regular red patches</td><td></td></tr><tr><th colspan="3">Heterozygotes</th></tr><tr><td><i>W</i></td><td>with any other allele</td><td>red</td></tr><tr><td><i>w^p</i></td><td>with either <i>w^s</i> or <i>w</i></td><td>white with regular red patches</td></tr><tr><td><i>w^sw</i></td><td></td><td>white stippled with red</td></tr></table> Arrange the alleles in a dominance hierarchy.	Homozygotes			<i>WW</i>	red		<i>ww</i>	pure white		<i>w^sw^s</i>	white stippled with red		<i>w^pw^p</i>	white with regular red patches		Heterozygotes			<i>W</i>	with any other allele	red	<i>w^p</i>	with either <i>w^s</i> or <i>w</i>	white with regular red patches	<i>w^sw</i>		white stippled with red	
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Q-II C	Present a mechanism for the following through pictorial diagram [Any Two]	8																
1	Competence factor mediated process of Bacterial transformation																	
2	F ⁺ to F ⁻ gene transfer																	
3	Affinity of “cro” and CI towards different operator sites																	
SECTION - II																		
Q-IIIA	Answer any two of the following	10																
1	With respect to the lambda phage genes comment about the fate (lytic/lysogenic) of phage lambda under the following conditions 1. High concentration of Cro : 2. Absence of CII 3. Mutated N Protein: 4. Mutated Q 5. No transcription from P _{RE}																	
2	Describe principle, method and application of RFLP																	
3	Short hair in rabbits (S) is dominant over long hair (s). The following crosses are carried out, producing the progeny shown. Give all possible genotypes of the parents in each cross. Parents Progeny (a) short _ short 4 short and 2 long (b) short _ short 8 short																	
Q-IIIB	Write a note on any two of the following:	10																
1	What is mutation? Define two broad types of mutation? Discuss the various types of chromosomal rearrangements with suitable pictorial presentation																	
2	What is somatic cell hybrid? Discuss the brief method for its generation. Discuss It's role in specifying the gene locations, provide a hypothetical example for the Same.																	
3	In <i>Neurospora</i>, a fungus with mating type A is crossed with a fungus that has mating type a, and the following linear octads are produced. <table><tr><td>A A A A a a a a</td><td>212</td></tr><tr><td>a a a a A A A A</td><td>221</td></tr><tr><td>A A a a A A a a</td><td>15</td></tr><tr><td>a a A A a a A A</td><td>17</td></tr><tr><td>A A a a a a A A</td><td>19</td></tr><tr><td>a a A A A A a a</td><td>16</td></tr><tr><td colspan="2"> </td></tr><tr><td>Total</td><td>500</td></tr></table> What is the genetic distance of the mating type locus to the centromere?	A A A A a a a a	212	a a a a A A A A	221	A A a a A A a a	15	a a A A a a A A	17	A A a a a a A A	19	a a A A A A a a	16			Total	500	
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Q-IIIC	Answer any one of the following:	5																
1	Write a note on heritability and its measurement																	
2	Discuss QTL mapping																	
Q-IIID	Answer any one of the following:	5																
1	Sex-limited and sex-influenced characters																	
2	Discuss extra nuclear inheritance																	